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Proposal Title

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Executive Summary

dabestr is the principal R implementation of estimation statistics (a data-analysis framework that reports effect sizes and confidence intervals in place of null-hypothesis significance testing). It is hosted on CRAN and downloadable from GitHub and implements methods published across two Nature Methods papers (Ho et al. 2019; Lu et al. 2026). Over successive releases it has grown from two-group mean differences into a feature-rich package spanning repeated measures, delta-delta (2×2) designs, mini-meta analysis, and binary proportions.

That growth has outpaced the package's original architecture. Dabestr contains first class statistical concepts wrapped in less than first-class software. They are encoded implicitly across the arguments of a single entry-point function, a set of design flags, and a heterogeneous result object that the plotting layer inspects directly. The result of such organic growth is a package that works but is increasingly hard to extend, test, and integrate with the rest of the R ecosystem.

We propose a focused modernization that makes the existing statistical grammar explicit: a documented, standardized, composable estimation-result object, with computation cleanly separated from visualization, validated against the current implementation as a frozen regression target, and released with full backward compatibility. Instead of new methodology, we want to consolidate the engineering, which keeps scientific risk low while producing reusable and reliable infrastructure that other R packages and visualization systems can build on.

Signatories

Project team

Contributors

Consulted

The Problem

Estimation statistics has become a mainstream alternative to significance testing across the life and behavioural sciences, reinforced by journal policy changes and a growing tool ecosystem. `dabestr` is the main way R users produce estimation analyses and plots. But the package exposes no stable, documented representation of an estimation result. A completed analysis is a large list carrying, simultaneously, raw data, comparison indices, control/test summaries, bootstrap and permutation results, design flags, and plotting configuration (labels, limits, colour mappings). The same object serves as analysis specification, results container, data carrier, design record, and plot configuration all at once.

Two structural consequences follow:

Data ingestion, experimental design, and analysis configuration are collapsed into one entry point. The `load()` function now encodes grouping structure, comparison structure (`idx`), paired/repeated-measures design, subject identifiers, proportional outcomes, mini-meta analysis, and delta–delta analysis. Each new method has historically been added as another argument, flag, or specialized code path rather than as an explicit component of a shared model. Visualization is coupled to internal statistical state. The plotting layer branches on whether an analysis is delta–delta or mini-meta, on the number and structure of comparison groups, and on other design-specific fields, then assembles the panels accordingly. Presentation logic therefore depends on knowledge of the internal object rather than on a documented contract.

Why is it a problem, and who does it affect?

Users cannot reliably program against results — extracting effect sizes, intervals, or bootstrap distributions for downstream reporting means reaching into undocumented internal structure that can change between releases. The maintainers face rising per-feature cost. Our own roadmap illustrates this directly: delta–delta alone has accumulated three separate open requests (simplify its input interface; support a “multiplex” form; support pooled variance between control and test groups). Each currently implies bespoke input handling and plotting logic. This is the accumulation pattern in miniature. The wider R ecosystem cannot easily consume estimation results. Because there is no standardized, tidy-friendly result object, alternative or `ggplot2`-native plotting layers, report-generation tools, and other statistical packages cannot build on `dabestr` output without depending on its internals.

What will solving it enable?

A documented estimation-result object turns `dabestr` from a self-contained analysis-and-plot tool into reusable infrastructure: results become ordinary, inspectable, composable R objects; visualization becomes one consumer of a public contract rather than a privileged insider; and future estimands, designs, resampling procedures, and plotting systems can plug into stable interfaces instead of adding specialized pathways. It also enables principled harmonization of concepts with DABEST-Python where that serves users, without forcing non-idiomatic R.

An example in-text citation ([Wickham 2016](#)).

The proposal

Overview

Detail

Minimum Viable Product

Architecture

Assumptions

External dependencies

Project plan

Start-up phase

Technical delivery

Other aspects

Budget & funding plan

Success

Definition of done

Measuring success

Future work

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York.
<https://ggplot2.tidyverse.org>.